Week 11: 9–13 November 2026
الأسبوع 11: من 9 - إلى 13 نوفمبر 2026

Subject / الموضوع: AP Chemistry **Unit Title / عنوان الوحدة: Unit 7 Equilibrium –** **Grade / الصف: 12**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Session 1 الحصة الأولى 7.5 Magnitude of the Equilibrium Constant	**DOK 3 – Explain:** I can analyze the magnitude of K to explain whether reactants or products are favored at equilibrium, and justify my predictions with particle-level reasoning.	3.2 Highlight big ideas, patterns, and critical features using organizers and examples. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 7.2 Make learning relevant, authentic, and connected to students' backgrounds.	Kindly check Toddle/Teams Folder for the lesson PPT and practice sheet.
Session 2 الحصة الثانية 7.6 Properties of the Equilibrium Constant	**DOK 3 – Analyze:** I can analyze how changing a chemical equation (reversing it, multiplying it by a factor, or adding equations) changes K, and justify the K for an overall multistep reaction.	3.3 Guide processing with explicit steps, chunking, and interactive models. 2.2 Clarify syntax/structure and highlight relationships in text, formulas, diagrams. 6.2 Support planning with templates, strategy prompts, and modeled thinking.	Kindly check Toddle/Teams Folder for the lesson PPT and practice sheet.
Session 3 الحصة الثالثة 7.7 Calculating Equilibrium Concentrations	**DOK 3 – Analyze:** I can use initial conditions and K to calculate equilibrium concentrations or partial pressures with an ICE table, and justify when the small-x approximation is valid.	5.3 Build fluency with graduated scaffolds and models. 8.2 Vary challenge level with differentiated tools, resources, and difficulty. 8.3 Foster collaboration and community with structured group roles and supports. **Formative:** ICE-table problems and mini-whiteboard checks	Kindly check Toddle/Teams Folder for the lesson PPT and practice sheet.
Session 4 الحصة الرابعة Lab: Determining an Equilibrium Constant	**DOK 3 – Investigate and evaluate:** I can collect and analyze experimental data to determine the equilibrium constant for a reaction, and evaluate how consistent my results are and the sources of error.	4.1 Provide alternatives for physical action and methods of response. 6.4 Support progress monitoring with rubrics, checklists, and visual progress tools. 9.3 Develop self-assessment and self-reflection with tools, charts, and feedback displays. **Formative:** lab data analysis and lab report	Kindly check Toddle/Teams Folder for the lesson PPT and practice sheet.

Homework **الواجبات**			
Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	This week students finish the equilibrium constant topics (7.5–7.7) and apply them in a lab, where they determine K from their own experimental data. Lab safety: students must wear goggles and follow the lab safety rules. Students need a calculator in every session.		

Week 11 الأسبوع 11

Technology Integration دمج التكنولوجيا	AP Classroom (AP Daily videos and Topic Questions), Toddle for lesson materials and the lab sheet, and a spreadsheet to process lab data.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics (exponents, solving for x in ICE tables and approximations) and Physics (measuring light absorbance and colour).
External Assessment Integration دمج التقييمات الخارجية	AP Chemistry Exam (College Board): Course and Exam Description Unit 7; AP-style lab skills for the free-response section.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Thinking (analysing data to calculate K), collaboration (working in lab groups), communication (writing a clear lab report) and self-management (working safely in the lab).
Standards المعايير	• **7.5 Magnitude of the Equilibrium Constant – TRA-7.C:** Explain the relationship between very large or very small values of K and the relative concentrations of chemical species at equilibrium. • **7.6 Properties of the Equilibrium Constant – TRA-7.D:** Represent a multistep process with an overall equilibrium expression, using the constituent K expressions for each individual reaction. • **7.7 Calculating Equilibrium Concentrations – TRA-7.E:** Identify the concentrations or partial pressures of chemical species at equilibrium based on the initial conditions and the equilibrium constant.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	**Inquiry:** How can we use K to predict, and then measure, what is in a reaction mixture at equilibrium? **TOK:** Each lab trial gives a slightly different K. When do differences in data count as evidence, and when are they just experimental error?
Success Criteria معايير النجاح **(Measurable)**	The student has met the learning targets by: لقد حقق الطالب/الطالبة أهداف التعلم من خلال: **Session 1 – 7.5 Magnitude of the Equilibrium Constant** • **MUST (DOK 1):** I can state that when K is much greater than 1 products are favored, and when K is much less than 1 reactants are favored. • **SHOULD (DOK 2):** I can rank several reactions by how far they proceed, using their K values. • **COULD (DOK 3):** I can justify, using K and concentration data or particle diagrams, why a reaction with a very large K is treated as going to completion. **Session 2 – 7.6 Properties of the Equilibrium Constant** • **MUST (DOK 1):** I can state that reversing an equation gives K = 1/K, and multiplying an equation by n gives K raised to the power n. • **SHOULD (DOK 2):** I can calculate K for an overall reaction by adding steps and multiplying their K values. • **COULD (DOK 3):** I can justify these rules from the K expressions and compare them with Hess's law (ΔH values are added, K values are multiplied). **Session 3 – 7.7 Calculating Equilibrium Concentrations**

	• **MUST (DOK 1):** I can set up an ICE table from the balanced equation and the initial concentrations. • **SHOULD (DOK 2):** I can solve for x and calculate all equilibrium concentrations or partial pressures. • **COULD (DOK 3):** I can justify when the small-x approximation is valid (the 5% rule) and check my answer by substituting back into the K expression. **Session 4 – Lab: Determining an Equilibrium Constant** • **MUST (DOK 1):** I can follow the lab procedure safely and record my data in a table with units. • **SHOULD (DOK 2):** I can use my experimental data in an ICE table to calculate K for each trial. • **COULD (DOK 3):** I can compare K values across trials, evaluate how consistent they are, and explain the sources of error and how to reduce them.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Conservation الحفاظ علي البيئة Global Understanding التفاهم العالمي Arabic Language اللغة العربية
	Area: المجال Sustainable industry and energy in the UAE, linked to the UAE Net Zero by 2050 strategic initiative.
	Implementation: التطبيق • Sessions 1–3: students use K to analyze the reversible reaction that makes ammonia, as produced in Ruwais, and calculate the equilibrium concentrations in the reaction mixture. • Session 4: students link the lab to how chemists in the UAE measure concentrations, such as testing the quality of drinking water from desalination plants. • Arabic key terms: ثابت الاتزان (equilibrium constant), حاصل التفاعل (reaction quotient), التركيز (concentration), الضغط الجزئي (partial pressure), الاتزان الكيميائي (chemical equilibrium).